

CLIENT

React Frontend

- Analysis
- Results
- Report
- Workspaces
- Sessions
- Settings

REVERSE PROXY

nginx

ORCHESTRATOR · FASTAPI

Agent

Iterative LLM loop
Multi-turn tool calls
Session context

Task Runner

Background thread pool
Inference queue
Error explanation

Watcher Service

Filesystem polling
Triggers agent on new file
SSE console stream

Tool Registry

MCP client · discovers and calls tools at runtime

REST API + SSE

query · events · tasks · sessions · upload · workspaces

SQLITE

sessions

messages

tasks

files

workspaces

LLM BACKEND

Gemini API

Ollama

MCP SERVER

monai

stdio

Analyze image
List models
Download model
Run inference

MONAI bundles

MCP SERVER

radlex

stdio

List subspecialties
Find templates
Generate report

MCP SERVER

utils

stdio

Parse DICOM
Write file
List directory
Find files

MCP SERVER

cellpose

stdio

Segment cells 2D
Segment cells 3D
Save overlay

MCP SERVER

ipath

stdio

Fetch thumbnail
Get dimensions
Scale ROI
Fetch ROI

iPath platform

MCP SERVER

fhir

HTTP

Patient records
FHIR standard